

AI-Based Student Dropout Prediction and Counseling System

13-Day Sprint Documentation

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Day 1 | June 01, 2025 | Project Title Finalization

Deliverable: Approved Project Title

1.1 Project Title

AI-Based Student Dropout Prediction and Counseling System

1.2 Project Overview

Educational institutions face a major challenge in identifying students who are at risk of dropping out before it is too late. Traditional monitoring methods rely heavily on manual observation and academic performance reviews, which often fail to detect early warning signs.

The proposed system uses Artificial Intelligence and Machine Learning techniques to predict the likelihood of student dropout by analysing academic, attendance, behavioural, and demographic factors. Based on the prediction results, the system provides personalized counseling recommendations to support at-risk students and improve retention rates.

1.3 Abstract

The AI-Based Student Dropout Prediction and Counseling System is a web-based intelligent platform developed to assist educational institutions in identifying students who may discontinue their studies. The system collects and analyses student-related data such as attendance records, academic performance, internal assessment scores, and behavioural indicators.

Using a machine learning model, the system predicts dropout risk levels and classifies students into different categories such as Low Risk, Moderate Risk, and High Risk. Counselors and administrators can view these predictions and provide timely interventions through counseling sessions and personalized recommendations.

The proposed solution aims to improve student retention, reduce dropout rates, and support evidence-based decision-making within educational institutions.

1.4 Scope of the Project

The scope of the proposed system includes:

- Student registration and authentication.
- Collection and storage of academic and attendance data.
- Prediction of dropout probability using machine learning algorithms.
- Identification of high-risk students.
- Counseling recommendation generation.
- Counselor management and intervention tracking.
- Dashboard and reporting facilities for administrators.

- Maintenance of prediction history.

Features such as SMS notifications and parent mobile applications are considered future enhancements and are not included in Version 1.0.

1.5 Expected Benefits

For Students

- Early identification of academic difficulties.
- Personalized counseling support.
- Improved academic guidance.

For Counselors

- Identification of students requiring immediate attention.
- Better counseling planning.
- Efficient tracking of intervention outcomes.

For Institutions

- Reduced dropout rates.
- Improved student retention.
- Data-driven decision making.

1.6 Deliverable

Approved Project Title: **AI-Based Student Dropout Prediction and Counseling System**

1.7 Remarks: Project title finalized successfully after reviewing the educational challenges associated with student retention and counseling support.

Day 2 | June 02, 2025 | Problem Statement & Requirement Gathering

Deliverable: Problem Statement

2.1 Problem Statement

Student dropout remains one of the major concerns faced by educational institutions worldwide. Various factors such as poor academic performance, irregular attendance, financial difficulties, lack of motivation, and personal issues contribute to students discontinuing their education.

Most institutions rely on manual monitoring systems, which are time-consuming and often incapable of identifying at-risk students at an early stage. As a result, intervention measures are delayed, reducing their effectiveness.

There is a need for an intelligent system capable of predicting dropout risk using historical and real-time student data while simultaneously recommending counseling strategies that support students before they decide to leave their education.

The proposed AI-Based Student Dropout Prediction and Counseling System addresses this challenge by combining machine learning prediction models with counseling support mechanisms.

2.2 Functional Requirements

- **FR-01: User Authentication:** The system shall allow Students, Counselors, and Administrators to securely register and log in using valid credentials.
- **FR-02: Student Data Management:** The system shall allow administrators to manage student records including academic performance, attendance, and personal information.
- **FR-03: Dropout Prediction:** The system shall analyse student information using a machine learning model and predict the probability of dropout.
- **FR-04: Risk Classification:** The system shall classify students into Low Risk, Moderate Risk, and High Risk categories based on prediction outcomes.
- **FR-05: Counseling Recommendation:** The system shall provide personalized counseling recommendations for students identified as being at risk.
- **FR-06: Counseling Session Tracking:** Counselors shall be able to record counseling sessions and monitor student progress over time.
- **FR-07: Report Generation:** Administrators shall generate reports containing prediction statistics, dropout trends, and counseling activities.

2.3 Non-Functional Requirements

ID	Requirement	Detail
NFR-01	Performance	The prediction result must be generated within 5 seconds under normal conditions.
NFR-02	Usability	The interface should be simple, responsive, and accessible on mobile and desktop devices.
NFR-03	Security	Passwords must be encrypted using bcrypt. JWT tokens must be used for authentication and authorization.
NFR-04	Reliability	The system should maintain 99% uptime during academic operations.
NFR-05	Scalability	The system should support at least 2,000 student records without performance degradation.
NFR-06	Maintainability	The machine learning model should be replaceable without affecting frontend functionality.
NFR-07	Accuracy	The prediction model should achieve a minimum validation accuracy of 90%.

2.4 Stakeholder Requirements

Stakeholder	Requirements	Priority
Students	Early identification of dropout risk and counseling recommendations	High
Counselors	Access to high-risk student lists and counseling history	High
Administrators	Student management, report generation, and system monitoring	High
College Management	Improved student retention and performance statistics	High
Project Evaluator	Well-documented GitHub repository, ER Diagram, APIs, and reports	High
System Administrator	Backup, maintenance, and user management facilities	Medium

2.5 Constraints

- The project must be completed within the academic semester timeline.
- Student data availability may be limited during development.

- Publicly available educational datasets will be used for ML training.
- The system will initially support English language only.
- SMS and parent notification features are excluded from Version 1.0.
- Deployment will be performed using free-tier cloud platforms during evaluation.

Remarks: Problem statement and requirements were finalized after analysing challenges related to student dropout and intervention practices.

Day 3 | June 03, 2025 | Objective Definition

Deliverable: Project Objectives

3.1 Primary Objective

To design and develop an intelligent web-based system capable of predicting student dropout risk using machine learning techniques and recommending timely counseling interventions to improve student retention rates.

3.2 Specific Objectives

- **Objective 1:** Collect and preprocess student-related datasets containing attendance records, academic performance indicators, demographic information, and historical dropout status.
- **Objective 2:** Train and evaluate machine learning classification models capable of predicting dropout risk with high accuracy.
- **Objective 3:** Develop a secure backend API using Spring Boot to process student information and communicate with the machine learning prediction engine.
- **Objective 4:** Design and develop a responsive React frontend for Students, Counselors, and Administrators.
- **Objective 5:** Create a normalized MySQL database schema capable of storing student records, prediction results, counseling sessions, and reports.
- **Objective 6:** Implement role-based authentication using JWT to ensure secure access for all system users.
- **Objective 7:** Maintain a structured GitHub repository containing source code, diagrams, documentation, and development evidence.

3.3 Objectives Mapped to Deliverables

Objective	Description	Targeted Day
O-1	Dataset collection and preprocessing	Day 1–3
O-2	Machine learning model development	Day 7
O-3	Backend API development	Day 5
O-4	Frontend interface design	Day 9–11
O-5	Database design	Day 6–8
O-6	Authentication implementation	Day 10
O-7	GitHub repository and documentation	Day 13

3.4 Expected Outcomes

- A functional web application capable of predicting student dropout risk.
- Early identification of students requiring counseling interventions.
- Improved student retention through data-driven recommendations.
- Secure management of academic and counseling information.
- A structured GitHub repository suitable for academic evaluation.
- Documentation sufficient for future enhancement and deployment.

3.6 Remarks: Project objectives were defined successfully and aligned with the expected deliverables of the development sprint.

Day 4 | June 04, 2025 | User & Module Identification

Deliverable: Module List

4.1 Identified Users

User Type 1: Student (Primary User)

The student is the primary beneficiary of the system. Students access the portal to view their academic progress, prediction status, and counseling recommendations. They can monitor their own risk category and seek support when necessary.

User Type 2: Counselor

Counselors play a vital role in monitoring students who are classified as moderate-risk or high-risk. They review

prediction outcomes, conduct counseling sessions, provide intervention strategies, and record observations for future reference.

User Type 3: Administrator

The administrator manages the overall functioning of the system. Responsibilities include maintaining student records, managing counselor accounts, monitoring system usage, generating reports, and supervising prediction outcomes.

4.2 System Modules

Mod No.	Module Name	What It Does	Who Uses It
M-01	Authentication Module	Handles registration, login, logout, JWT token generation, and role validation.	Student, Counselor, Admin
M-02	Student Management Module	Stores and manages student profiles, attendance records, and academic information.	Admin
M-03	Dropout Prediction Module	Processes student data using the ML model and predicts dropout risk levels.	System
M-04	Counseling Recommendation Module	Provides personalized counseling suggestions for at-risk students.	Student, Counselor
M-05	Counseling Session Module	Allows counselors to record and monitor counseling activities.	Counselor
M-06	Dashboard & Reporting Module	Displays analytics, prediction summaries, and institutional reports.	Admin
M-07	Notification Module	Sends alerts and notifications regarding counseling sessions and interventions.	Admin, Student

4.3 Module Interaction Summary

When a student logs into the system (M-01), they are redirected to their dashboard where they can view prediction status and counseling recommendations (M-04). Administrators maintain student records (M-02), which are processed by the prediction engine (M-03). Students identified as being at risk are assigned to counselors through the counseling session module (M-05). Institutional reports are generated through the reporting module (M-06), while notifications (M-07) keep users informed about important updates.

4.5 Remarks: System users and modules were identified successfully with clearly defined responsibilities.

Day 5 | June 05, 2025 | Use Case Diagram Preparation & CRUD APIs

Deliverable: CRUD APIs

5.1 Actors

Student: The primary actor who accesses predictions and counseling recommendations.

Counselor: The secondary actor responsible for conducting counseling sessions and monitoring student progress.

Administrator: The actor responsible for system administration and report generation.

AI Prediction Engine: A system actor that analyses student data and generates dropout predictions.

5.2 Use Cases Summary

- **Student:** Register Account, Login, View Prediction, View Recommendations, View Counseling History, Logout.
- **Counselor:** Counselor Login, View High-Risk Students, Record Counseling Session, Update Counseling Notes.
- **Administrator:** Admin Login, Manage Students, Manage Counselors, Generate Reports, Monitor Predictions.

5.3 CRUD API Endpoints

Authentication APIs

Method	Endpoint	Action	Auth Required
POST	/api/auth/register	Register student	No
POST	/api/auth/login	Login user	No
GET	/api/auth/profile	View profile	Yes
PUT	/api/auth/profile	Update profile	Yes
DELETE	/api/auth/account	Delete account	Yes

Prediction APIs

Method	Endpoint	Action	Auth Required
POST	/api/predictions	Generate prediction	Yes
GET	/api/predictions/:id	View prediction	Yes
GET	/api/predictions/history	Prediction history	Yes
DELETE	/api/predictions/:id	Delete prediction	Yes

Counseling APIs

Method	Endpoint	Action	Auth Required
POST	/api/counseling	Create counseling record	Counselor
GET	/api/counseling	View counseling records	Yes
PUT	/api/counseling/:id	Update counseling notes	Counselor
DELETE	/api/counseling/:id	Delete counseling record	Counselor

Student APIs (Admin Managed)

Method	Endpoint	Action	Auth Required
GET	/api/students	View students	Admin
POST	/api/students	Add student	Admin
PUT	/api/students/:id	Update student	Admin
DELETE	/api/students/:id	Delete student	Admin

5.6 Remarks: Use cases and CRUD APIs were prepared successfully following REST architectural principles.

Day 6 | June 06, 2025 | Database Requirement Analysis

Deliverable: Table List

6.1 Relational Storage Strategy

The proposed system requires persistent storage to maintain student information, prediction outcomes, counseling activities, and institutional reports. A relational database (MySQL) is selected because the information is highly structured and involves multiple relationships among entities.

6.2 Identified Tables & Relationships

Table Name	Stores	Key Relationships
users	Authentication details	Referenced by students and counselors
students	Student information	Referenced by predictions
counselors	Counselor details	Referenced by counseling_sessions
predictions	Dropout predictions	Belongs to students
counseling_sessions	Counseling activities	Links students and counselors
recommendations	Counseling suggestions	References predictions
reports	Institutional reports	Generated by administrators
notifications	Alerts and reminders	Belongs to users

6.3 Data Volume Estimates (Year 1)

- **students:** 1,000–5,000 rows (Growth: Annual admissions)
- **predictions:** 3,000–10,000 rows (Growth: Semester-based processing)
- **counseling_sessions:** 1,000–5,000 rows (Growth: Event-driven by risk thresholds)

6.6 Remarks: Database requirements were analysed successfully and the required tables were identified with clear relationships.

Day 7 | June 07, 2025 | ER Diagram Design

Deliverable: ER Diagram Mapping

7.1 Entity Specifications

The ER Diagram layout models the explicit structural attributes for each entity. Normalization constraints have been checked to ensure zero redundancy.

Entity Attribute Schemes

- **Users:** `user_id` (PK), `email`, `password`, `role`, `created_at`
- **Students:** `student_id` (PK), `register_number`, `student_name`, `department`, `year`, `attendance_percentage`, `cgpa`, `user_id` (FK)
- **Counselors:** `counselor_id` (PK), `counselor_name`, `specialization`, `contact_number`, `user_id` (FK)
- **Predictions:** `prediction_id` (PK), `dropout_probability`, `risk_level`, `prediction_date`, `student_id` (FK)
- **Counseling Sessions:** `session_id` (PK), `session_date`, `notes`, `status`, `student_id` (FK), `counselor_id` (FK)
- **Recommendations:** `recommendation_id` (PK), `recommendation_text`, `prediction_id` (FK)

7.2 Cardinality & Relationships

Parent Entity	Cardinality Mapping	Child Entity
Users	1 : 1	Students
Users	1 : 1	Counselors
Students	1 : M	Predictions
Students	1 : M	Counseling Sessions
Counselors	1 : M	Counseling Sessions
Predictions	1 : M	Recommendations

7.6 Remarks: ER Diagram logic prepared successfully and reviewed for normalization and relationship consistency.

Day 8 | June 08, 2025 | SQL Schema Creation

Deliverable: SQL Schema DDL

8.1 DDL Statements

```
CREATE TABLE users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    email VARCHAR(100) UNIQUE,
    password VARCHAR(255),
    role VARCHAR(20),
```

```

        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    );

CREATE TABLE students (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    register_number VARCHAR(20),
    student_name VARCHAR(100),
    department VARCHAR(50),
    year INT,
    attendance_percentage DECIMAL(5,2),
    cgpa DECIMAL(3,2),
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES users(user_id)
);

CREATE TABLE counselors (
    counselor_id INT PRIMARY KEY AUTO_INCREMENT,
    counselor_name VARCHAR(100),
    specialization VARCHAR(100),
    contact_number VARCHAR(15),
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES users(user_id)
);

CREATE TABLE predictions (
    prediction_id INT PRIMARY KEY AUTO_INCREMENT,
    dropout_probability DECIMAL(5,2),
    risk_level VARCHAR(20),
    prediction_date DATE,
    student_id INT,
    FOREIGN KEY (student_id) REFERENCES students(student_id)
);

CREATE TABLE counseling_sessions (
    session_id INT PRIMARY KEY AUTO_INCREMENT,
    session_date DATE,
    notes TEXT,
    status VARCHAR(50),
    student_id INT,
    counselor_id INT,
    FOREIGN KEY (student_id) REFERENCES students(student_id),
    FOREIGN KEY (counselor_id) REFERENCES counselors(counselor_id)
);

```

8.5 Remarks: SQL Schema created successfully with normalized table structures and explicit foreign key rules.

Day 9 | June 09, 2025 | UI Wireframe Design

Deliverable: Page Layouts

9.1 Structural Wireframe Mapping

- **Login Screen:** Layout contains multi-role selectors, authentication input text-fields, and structured error boundaries.
- **Student Dashboard:** Visual indicators for risk probability scores and dynamic textual counseling advice components.
- **Counselor Interface:** Tables highlighting cross-referenced high-risk metrics, priority filtering, and history notes.
- **Admin Analytical Desk:** Contains structural metrics for high-risk cohorts, enrollment trends, and automated CSV report extractors.

9.10 Remarks: UI wireframes designed successfully to guide frontend architectural component boundaries.

Day 10 | June 10, 2025 | Login & Dashboard UI Design

Deliverable: UI Screens

10.1 UI High-Fidelity Specifications

The transformation from basic structural wireframes into concrete high-fidelity interfaces ensures streamlined workflows across roles:

- **Login Interface:** Enforces clean form controls with real-time UI state feedback and conditional validation.
- **Student Views:** Clean graphical widgets highlighting current risk classifications (Red for High, Amber for Moderate, Green for Low risk levels).
- **Counselor Framework:** Prioritizes workflow automation with instant scheduling dialogs and rich-text clinical inputs.

10.7 Remarks: Login and Dashboard interfaces were designed successfully for all system roles.

Day 11 | June 11, 2025 | Navigation & Form Design

Deliverable: UI Prototype Navigation Flow

11.1 Flow Schemes

Student Path: Login → Dashboard View → Prediction Details → Recommendations Panel → Session History List.

Counselor Path: Login → Case Portfolio → High-Risk Filtering → Session Form Execution → Submit Notes.

11.2 UI Form Input Boundary Rules

Field Name	Validation Constraint Rule
Email Field	Must match exact RFC-compliant organizational email pattern
Password Pattern	Minimum 8 characters length string containing alpha-numeric notation
Attendance Percentage	Strict numerical domain boundary value [0.00 to 100.00]
CGPA Metric	Precision floating range validation check [0.00 to 10.00]

11.7 Remarks: Navigation structures and user forms were finalized successfully.

Day 12 | June 12, 2025 | Design Review

Deliverable: Design Approval Sign-Off

12.1 Refinements & Quality Resolutions

- **Identified Gap 1:** Weak registration parameters missing explicit data-type guards.
Resolution: Extended form-schema layers with customized validation boundaries.
- **Identified Gap 2:** General admin dashboards lacked micro-level department categorizations.
Resolution: Added granular department filtering criteria to data charts.

12.6 Remarks: Architectural and design review complete. System blueprint approved for technical development phase.

Day 13 | June 13, 2025 | Frontend Environment Setup

Deliverable: React Project Setup

13.1 Technical CLI Scaffolding Steps

```
# Execute project generation using Vite engine
npm create vite@latest student-dropout-prediction -- --template react
cd student-dropout-prediction
npm install

# Inject atomic styling capabilities via Tailwind CSS
npm install -D tailwindcss postcss autoprefixer
npx tailwindcss init -p

# Incorporate core routing modules and HTTP client engine
npm install react-router-dom axios
```

13.2 Scaffolded Architecture Structure

```
student-dropout-prediction/
├── src/
│   ├── components/      # Navbar.jsx, PredictionCard.jsx, StudentForm.jsx
│   ├── pages/           # Login.jsx, StudentDashboard.jsx, AdminDashboard.jsx
│   ├── services/        # api.js (Axios base client interceptors configured)
│   └── context/         # AuthContext.jsx (JWT structural state handler)
├── .env                 # VITE_API_URL=http://localhost:8080
└── package.json
```

13.3 Sprint Completion Milestone Metrics

Sprint Phase	Timeline Window	Execution Status
Project Scope Definition	Day 1 – Day 3	100% Completed
System Design Phase	Day 4 – Day 5	100% Completed
Database Relational Modeling	Day 6 – Day 8	100% Completed
UI/UX High Fidelity Prototyping	Day 9 – Day 11	100% Completed
Code Scaffolding & Review	Day 12 – Day 13	100% Completed

Sprint Outcome Note: Planning, database schema declarations, and frontend environmental setups have successfully concluded. The system architecture is primed for feature-level development sprints.